AI-Based Hyper-Local Aviation Weather Forecast Verification System for Sri Lankan Monsoon Patterns

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Project Proposal Report

Sankadeepa W.D.I – IT22113986

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Sankadeepa W.D.I	**IT22113986**	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Ms. Shashika Lokuliyana		

ABSTRACT

Weather forecasting plays a critical role in aviation safety and operational decision-making. Pilots rely on aviation weather reports such as Terminal Aerodrome Forecast (TAF) and METAR to obtain atmospheric information before aircraft take-off and landing. These reports provide important meteorological parameters including temperature, wind speed, wind direction, atmospheric pressure, and humidity, which directly influence aircraft performance and airport operations. However, existing aviation weather forecasting systems primarily depend on global numerical weather prediction models such as the Global Forecast System (GFS) and the European Centre for Medium-Range Weather Forecasts (ECMWF). While these models provide reliable large-scale forecasts, they often fail to capture localized atmospheric variations occurring at specific airport terminals. Sri Lanka experiences complex weather conditions influenced by monsoon circulation systems including the Southwest Monsoon, Northeast Monsoon, and two inter-monsoon periods. These seasonal patterns frequently produce rapid weather changes that affect aviation operations. Existing forecasting systems rarely consider hyper-local monsoon-driven atmospheric interactions between temperature, humidity, and pressure.

This research proposes an AI-based hyper-local aviation weather forecasting and verification system that utilizes airport-level meteorological data to generate short-term weather predictions. The proposed solution includes a pressure-humidity-temperature coupled prediction model for the Northeast Monsoon and a weather visualization module that presents validated weather predictions using temperature heatmaps and wind pattern maps. Machine learning techniques including time-series forecasting models will be used to analyze historical weather datasets collected from Bandaranaike International Airport. The system will compare predicted results with official aviation weather reports such as METAR and TAF to evaluate prediction reliability. The expected outcome of the research is an intelligent aviation weather guidance system capable of improving localized weather prediction accuracy and supporting safer flight operations. The system contributes to improved aviation safety and efficient transportation infrastructure in Sri Lanka.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ML	Machine Learning
TAF	Terminal Aerodrome Forecast
METAR	Meteorological Aerodrome Report
GFS	Global Forecast System
ECMWF	European Centre for Medium-Range Weather Forecasts
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
QNH	Atmospheric Pressure at Mean Sea Level
SDG	Sustainable Development Goal
UI	User Interface
WBS	Work Breakdown Structure

Table 1- List of Abbreviations

1. INTRODUCTION

Weather forecasting plays a critical role in aviation safety and operational decision-making. Pilots and aviation forecasters rely on aviation weather reports such as the Terminal Aerodrome Forecast (TAF) and Meteorological Aerodrome Report (METAR) to understand atmospheric conditions before aircraft take-off and landing. These reports provide essential meteorological parameters including temperature, wind speed, wind direction, atmospheric pressure, and humidity, which directly influence aircraft performance, runway conditions, and flight safety. Accurate short-term weather prediction is therefore essential for supporting safe and efficient aviation operations.

Sri Lanka experiences complex weather patterns influenced by seasonal monsoon systems, particularly the Southwest Monsoon, Northeast Monsoon, and two inter-monsoon periods. During the Northeast Monsoon, rapid atmospheric changes frequently occur due to interactions between multiple meteorological variables such as temperature, humidity, and atmospheric pressure. These interactions often create localized atmospheric variations that can significantly affect aviation operations at airport environments such as Bandaranaike International Airport.

Most existing weather forecasting systems rely on large-scale numerical weather prediction models such as the Global Forecast System (GFS) and the European Centre for Medium-Range Weather Forecasts (ECMWF). While these systems provide reliable regional and global weather forecasts, they often struggle to capture hyper-local atmospheric variations occurring at individual airport terminals. In addition, many forecasting models analyze atmospheric variables independently, without fully considering the coupled relationships between temperature, humidity, and pressure that influence local weather dynamics.

To address this limitation, this research focuses on developing a pressure–humidity–temperature coupled prediction model specifically designed for Northeast Monsoon conditions in Sri Lanka. By analyzing the combined interaction between these atmospheric variables using machine learning techniques and airport-level meteorological data, the proposed model aims to improve the accuracy of short-term aviation weather prediction. The developed model will support more reliable localized weather forecasting and assist aviation forecasters in understanding atmospheric behavior during monsoon-driven weather conditions.

1.1 Background and Context

Weather forecasting plays a critical role in aviation safety and operational decision-making. Accurate atmospheric information is essential for aircraft take-off, landing, and route planning. Pilots and air traffic controllers rely on aviation weather reports such as the Terminal Aerodrome Forecast (TAF) and Meteorological Aerodrome Report (METAR) to obtain key meteorological parameters including temperature, wind speed, wind direction, atmospheric pressure, humidity, and visibility. These parameters directly influence aircraft performance, runway conditions, and flight safety.

The aviation industry is highly sensitive to weather conditions. Severe weather events such as thunderstorms, turbulence, strong winds, and lightning can significantly disrupt flight operations and pose safety risks. Studies in aviation meteorology indicate that adverse weather conditions contribute to a considerable proportion of aviation incidents and operational disruptions worldwide [5]. In tropical regions such as South Asia, convective weather systems are particularly common and are considered one of the most significant aviation weather hazards affecting flight operations [3]. In Sri Lanka, Bandaranaike International Airport frequently experiences lightning activity and convective weather conditions, especially during monsoon seasons, which can affect aircraft movements and airport operations [4].

Sri Lanka experiences complex meteorological conditions due to seasonal monsoon circulation systems, including the Southwest Monsoon, Northeast Monsoon, and two inter-monsoon periods. These seasonal systems produce rapid atmospheric variations such as sudden wind shifts, humidity changes, and pressure fluctuations. Previous studies have demonstrated that Artificial Neural Networks can be applied to forecast monsoon rainfall patterns in Sri Lanka by analyzing historical meteorological data [1]. Similarly, machine learning-based weather prediction models have been developed to improve short-term weather forecasting accuracy using historical weather datasets [2]. However, predicting localized weather variations in tropical environments remains a challenging problem.

Currently, most aviation weather forecasting systems rely on global numerical weather prediction models such as the Global Forecast System (GFS) and the European Centre for Medium-Range Weather Forecasts (ECMWF). These models simulate atmospheric processes using large-scale meteorological data to generate regional and global forecasts. Although these systems provide reliable large-scale weather predictions, they often struggle to accurately capture localized atmospheric variations occurring at specific airport terminals [9]. As a result, local weather changes that affect flight operations may not always be predicted accurately.

Recent advancements in artificial intelligence and deep learning have introduced new approaches for improving weather forecasting systems. AI-based frameworks have been proposed for automated aviation forecasting systems capable of generating terminal aerodrome forecasts using machine learning techniques [6]. In addition, deep learning architectures such as Convolutional Long Short-Term Memory (ConvLSTM) networks have

been used for precipitation nowcasting and short-term weather prediction by capturing spatial and temporal atmospheric patterns [7].

Furthermore, deep learning models have been widely studied for analyzing complex meteorological datasets and improving weather prediction accuracy [8]. Despite these advancements, many existing forecasting systems focus mainly on large-scale weather prediction rather than hyper-local forecasting for specific airport environments. To address these limitations, this research proposes an AI-based hyper-local aviation weather forecasting and verification system for Sri Lankan monsoon environments. The proposed system aims to analyze airport-level meteorological datasets and develop machine learning models capable of predicting key atmospheric parameters including temperature, humidity, wind conditions, and atmospheric pressure. In addition, the system incorporates a weather visualization component that generates graphical representations such as temperature heatmaps and wind pattern maps to support aviation decision-making. The proposed research contributes to the United Nations Sustainable Development Goals (SDGs), particularly SDG 9: Industry, Innovation and Infrastructure, by introducing technological innovation for improving aviation infrastructure systems, and SDG 13: Climate Action, which focuses on enhancing climate monitoring and weather prediction capabilities. Furthermore, this research belongs to the Artificial Intelligence and Data Science research cluster, specifically within the domain of intelligent aviation meteorology and weather forecasting systems.

1.2 Literature Review

Weather forecasting plays a critical role in aviation safety and operational decision-making. Pilots and aviation forecasters rely on aviation weather reports such as the Terminal Aerodrome Forecast (TAF) and Meteorological Aerodrome Report (METAR) to understand atmospheric conditions before aircraft take-off and landing. These reports provide essential meteorological parameters including temperature, wind speed, wind direction, atmospheric pressure, and humidity, which directly influence aircraft performance, runway conditions, and flight safety. Accurate short-term weather prediction is therefore essential for supporting safe and efficient aviation operations.

Sri Lanka experiences complex weather patterns influenced by seasonal monsoon systems, particularly the Southwest Monsoon, Northeast Monsoon, and two inter-monsoon periods. During the Northeast Monsoon, rapid atmospheric changes frequently occur due to interactions between multiple meteorological variables such as temperature, humidity, and atmospheric pressure. These interactions often create localized atmospheric variations that can significantly affect aviation operations at airport environments such as Bandaranaike International Airport.

Most existing weather forecasting systems rely on large-scale numerical weather prediction models such as the Global Forecast System (GFS) and the European Centre for Medium-Range Weather Forecasts (ECMWF). While these systems provide reliable regional and global weather forecasts, they often struggle to capture hyper-local atmospheric variations

occurring at individual airport terminals. In addition, many forecasting models analyze atmospheric variables independently, without fully considering the coupled relationships between temperature, humidity, and pressure that influence local weather dynamics.

To address this limitation, this research focuses on developing a pressure–humidity–temperature coupled prediction model specifically designed for Northeast Monsoon conditions in Sri Lanka. By analyzing the combined interaction between these atmospheric variables using machine learning techniques and airport-level meteorological data, the proposed model aims to improve the accuracy of short-term aviation weather prediction. The developed model will support more reliable localized weather forecasting and assist aviation forecasters in understanding atmospheric behavior during monsoon-driven weather conditions.

1.3 Research Gap

Although global weather forecasting models and aviation forecasting systems are widely used in aviation operations, most of these systems are developed using large-scale global meteorological datasets and are not specifically trained using local airport-level observations. As a result, these models often fail to accurately capture localized atmospheric variations that occur in specific airport environments such as Bandaranaike International Airport in Sri Lanka. In tropical regions influenced by monsoon circulation systems, rapid atmospheric changes frequently occur due to interactions between temperature, humidity, and atmospheric pressure. However, many existing forecasting approaches analyze these atmospheric variables independently, which may limit the ability of forecasting systems to represent real-world atmospheric behavior accurately.

Therefore, there is a need to develop a forecasting approach that can model the coupled relationship between temperature, humidity, and atmospheric pressure using local meteorological data. This research addresses this gap by developing a pressure–humidity–temperature coupled prediction model using machine learning techniques and airport-level meteorological datasets collected from Bandaranaike International Airport. By learning the interaction between these atmospheric variables during Northeast Monsoon conditions, the proposed model aims to improve the accuracy of short-term aviation weather prediction and provide more reliable weather guidance for aviation forecasters and pilots.

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to develop a hyper-local aviation weather forecasting and verification system using Artificial Intelligence (AI) and Machine Learning (ML) techniques to improve the reliability of short-term weather predictions for airport environments. The study focuses on analyzing meteorological data collected from Bandaranaike International Airport to identify local atmospheric patterns influenced by Sri Lanka’s monsoon systems.

The proposed system aims to develop a coupled atmospheric prediction model that learns the relationship between pressure, humidity, and temperature instead of predicting these variables independently. By modeling the interaction between these atmospheric parameters, the system aims to generate more accurate short-term aviation weather predictions that can support aviation operations.

In addition, the research will develop visualization mechanisms to present validated weather predictions through graphical representations such as temperature heatmaps and wind pattern maps. The system will also incorporate forecast verification by comparing AI-generated predictions with existing aviation weather reports such as METAR and Terminal Aerodrome Forecast (TAF). Overall, the proposed system aims to improve aviation decision-making and enhance flight safety in monsoon-influenced tropical airport environments.

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2.2 Specific Objectives

Analyze the relationship between pressure, humidity, and temperature during the Northeast Monsoon in order to understand their combined influence on local atmospheric behavior.

Develop a multi-variable machine learning model capable of predicting short-term aviation weather conditions by learning the coupled relationship between pressure, humidity, and temperature.

Evaluate the accuracy of the developed prediction model by comparing its results with existing aviation weather forecasts such as METAR observations and Terminal Aerodrome Forecast (TAF) reports.

Design weather visualization tools such as temperature heatmaps and wind pattern maps to present validated weather predictions in a format that can be easily interpreted by aviation forecasters and pilots.

3. METHODOLOGY

This research follows a Design Science research approach, which focuses on designing, developing, and evaluating a technological solution to address a real-world problem. In this study, an AI-based hyper-local aviation weather forecasting and verification system will be developed to improve short-term aviation weather prediction using meteorological data from Bandaranaike International Airport. The system aims to analyze the coupled relationship between atmospheric variables such as pressure, humidity, and temperature during the Northeast Monsoon season.

To achieve this objective, the research will apply machine learning techniques for time-series weather prediction. Historical meteorological data will first be collected and preprocessed to remove missing values and ensure data consistency. After preprocessing, feature engineering will be performed to analyze relationships between atmospheric variables. A **multi-variable machine learning model** will then be developed to learn the interactions between pressure, humidity, and temperature instead of predicting these parameters independently. Possible algorithms that will be explored include **Multivariate Linear Regression, Random Forest Regression, and Long Short-Term Memory (LSTM) networks**, which are commonly used for time-series forecasting and weather prediction.

The system will be implemented using Python-based data science tools and machine learning frameworks. Libraries such as Pandas and NumPy will be used for data preprocessing and analysis, while Scikit-learn and TensorFlow will be used for machine learning model development. Visualization tools such as Matplotlib and Seaborn will be used to generate graphical representations of weather data including temperature heatmaps and wind pattern maps. These tools are widely used in data science and provide efficient capabilities for handling large meteorological datasets and building predictive models.

Meteorological data used in this research will be collected from **Bandaranaike International Airport weather observations**, including parameters such as temperature, humidity, atmospheric pressure (QNH), wind speed, and wind direction. The dataset will consist of historical airport-level weather observations collected over multiple years to capture seasonal monsoon patterns. Since the research only uses environmental meteorological data and does not involve human participants or personal information, ethical risks are minimal. However, all data will be used responsibly and only for academic research purposes.

To validate the proposed prediction model, the system will compare AI-generated predictions with official aviation weather reports such as **METAR observations and Terminal Aerodrome Forecast (TAF) reports**. The accuracy of the model will be evaluated using common performance metrics including Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and prediction accuracy scores. These evaluation methods will help determine the effectiveness of the developed system in improving short-term aviation we

ather prediction and supporting decision-making for aviation forecasters and pilots.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed system focuses on developing a **coupled atmospheric prediction model** to improve short-term aviation weather forecasting during the Northeast Monsoon using meteorological data from Bandaranaike International Airport. Historical weather data including temperature, humidity, and atmospheric pressure will be collected and preprocessed as time-series data to train a multi-variable machine learning model. Unlike traditional forecasting approaches that predict atmospheric parameters independently, the proposed model learns the **interdependent relationship between pressure, humidity, and temperature**, enabling more accurate short-term aviation weather predictions. The predicted weather conditions will be evaluated through forecast verification by comparing model outputs with existing aviation weather reports. In addition, a weather visualization component will be developed to present validated predictions using graphical representations such as temperature heatmaps and wind pattern maps. These visualizations will assist aviation forecasters and pilots in easily understanding atmospheric conditions before flight operations. The modular design of the system allows future integration of additional atmospheric variables and datasets while maintaining reliable data management practices for meteorological information.

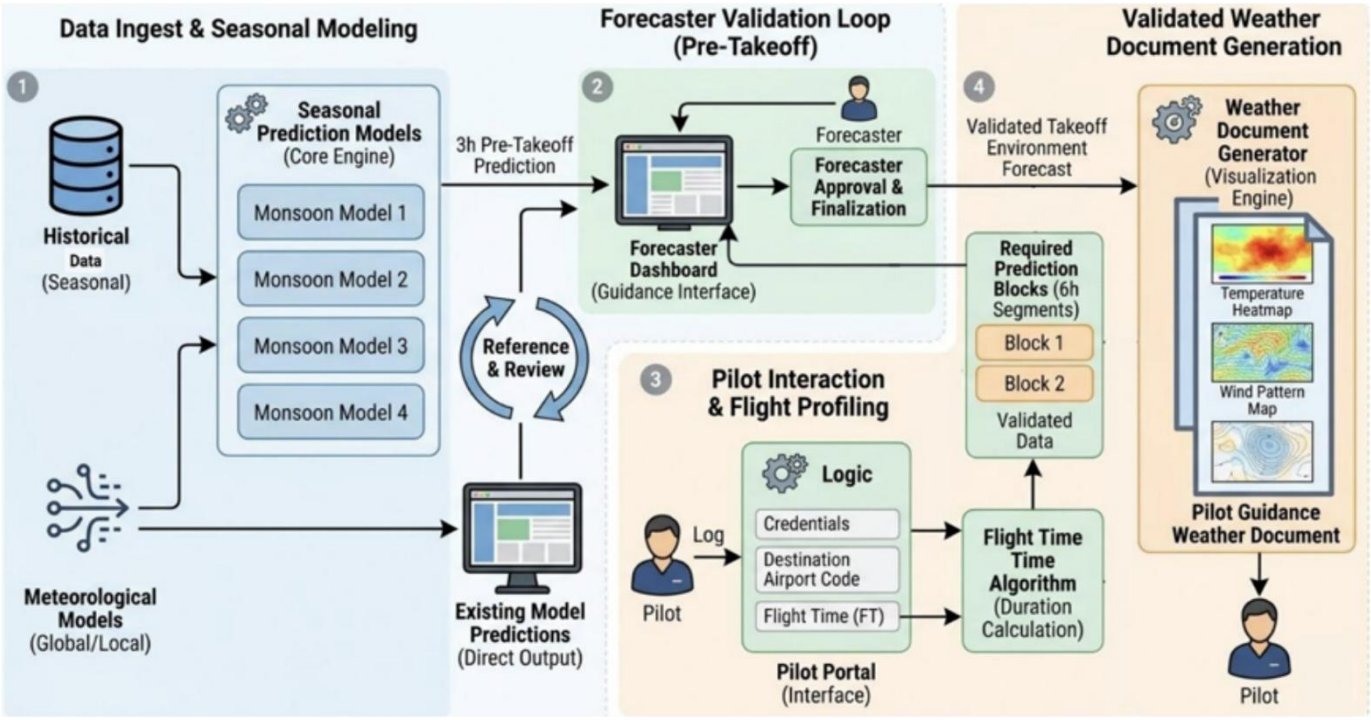


Figure 1 - System Diagram

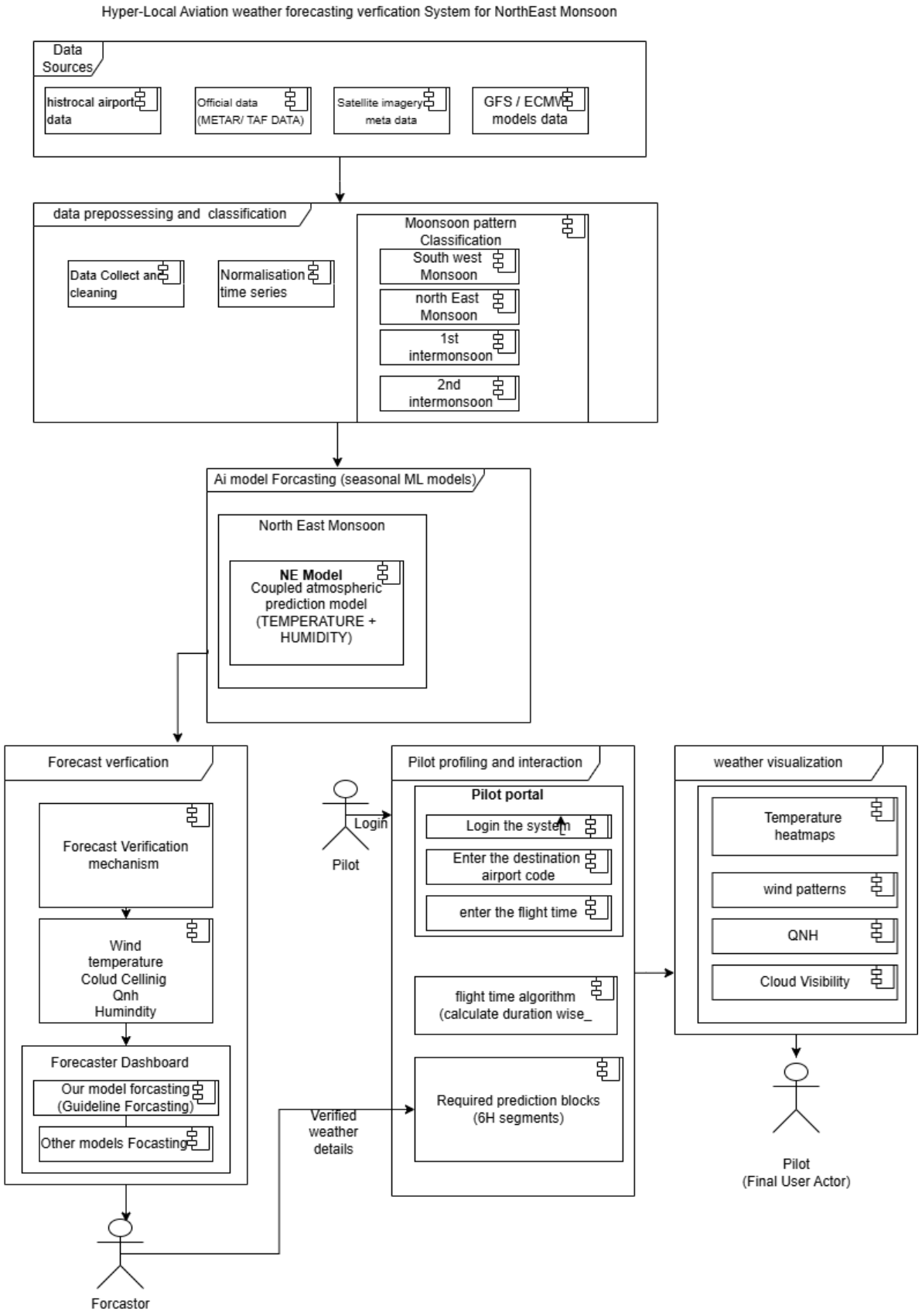


Figure 2- System Structure Diagram

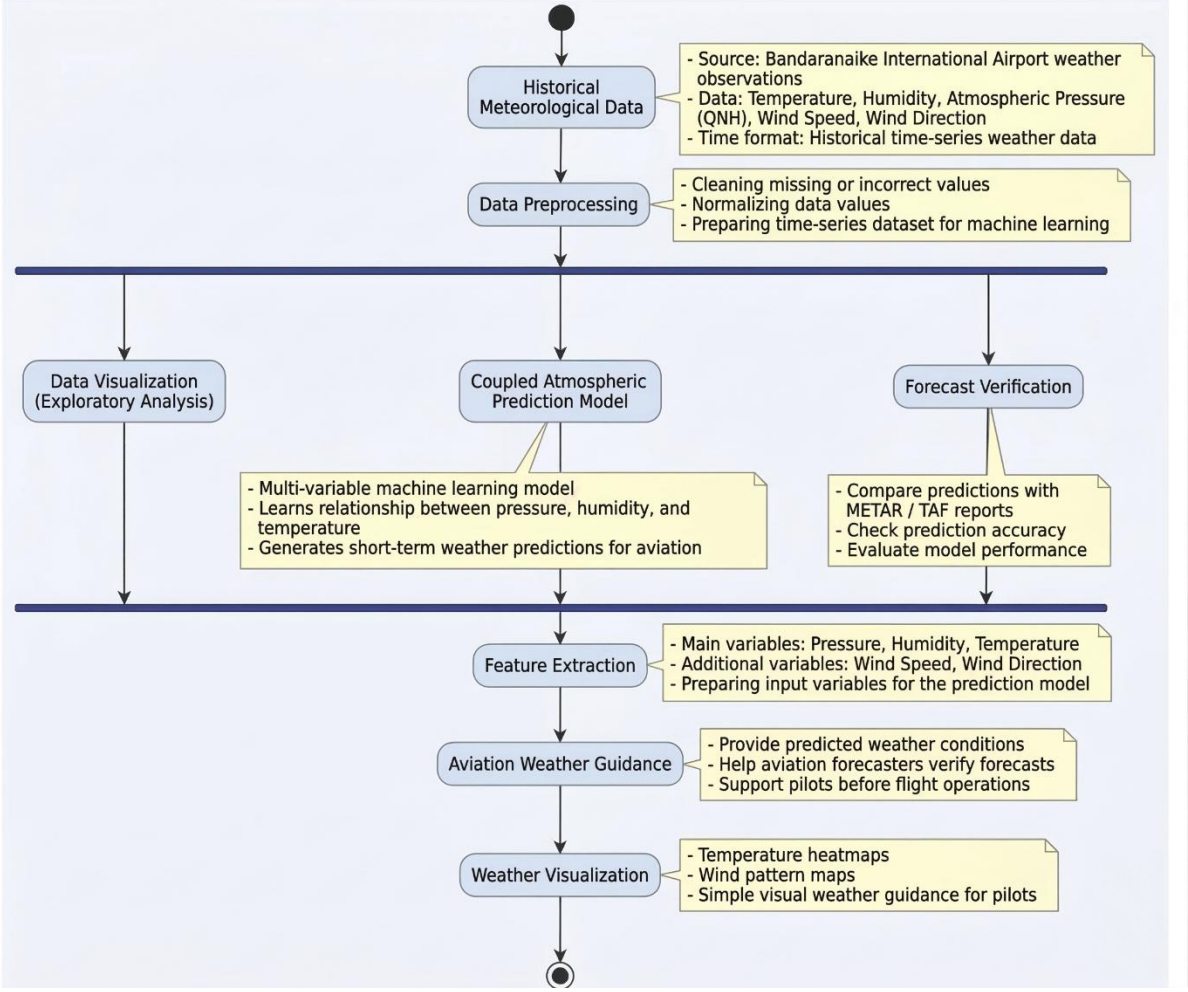


Figure 3 - Individual Component Workflow Diagram

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The primary stakeholders of the proposed aviation weather forecasting and verification system include:

- **Aviation Forecasters** – Meteorologists responsible for analyzing weather conditions and generating aviation weather reports. The system helps them verify forecast accuracy and analyze local atmospheric patterns.

- **Pilots** – Flight operators who rely on accurate weather information before aircraft take-off and landing. The system provides simplified weather visualizations to help pilots understand weather conditions.
- **Airport Operations Management** – Airport authorities responsible for ensuring safe and efficient airport operations. The system supports operational planning during adverse weather conditions.
- **Meteorological Departments** – Organizations responsible for collecting and analyzing weather data. The system supports improved interpretation of meteorological observations.

5.2 Functional Requirements

The system shall collect historical meteorological data from Bandaranaike International Airport.

The system shall preprocess meteorological datasets by cleaning, normalizing, and structuring the data for time-series analysis.

The system shall analyze the relationship between pressure, humidity, and temperature using a coupled machine learning prediction model.

The system shall generate short-term aviation weather predictions based on the trained machine learning model.

The system shall perform forecast verification by comparing model predictions with official aviation weather reports such as METAR and TAF.

The system shall generate visual weather representations including temperature heatmaps and wind pattern maps.

The system shall provide weather guidance outputs that help aviation forecasters and pilots interpret atmospheric conditions.

5.3 Non Functional Requirements

- **Performance** – The system should process meteorological datasets efficiently and generate predictions within a reasonable time for operational use.
- **Accuracy** – The machine learning model should provide reliable predictions by learning the relationship between pressure, humidity, and temperature.
- **Usability** – The visualization outputs should be easy to understand for aviation forecasters and pilots.
- **Scalability** – The system should allow integration of additional weather parameters or datasets from other airports in the future.

- **Security and Data Integrity** – Meteorological datasets should be securely stored and managed to ensure data reliability.
- **Reliability** – The system should consistently produce stable and interpretable weather predictions.

6. COMMERCIALIZATION PLAN

The proposed AI-based hyper-local aviation weather forecasting and verification system is developed mainly as a research prototype to demonstrate the use of machine learning techniques in aviation meteorology. The system focuses on analyzing local atmospheric conditions and improving short-term aviation weather prediction using meteorological data from Bandaranaike International Airport.

This project is conducted for academic research purposes and is related to a government domain. Therefore, the developed system is not intended to be commercially sold or used for revenue generation. Business models such as pricing strategies or subscription-based services are not considered within the scope of this study.

Instead, the main purpose of the system is to contribute to public safety and social benefit by supporting more reliable aviation weather forecasting. Accurate weather predictions can help aviation authorities and pilots make safer operational decisions, reducing weather-related risks in flight operations. Therefore, the research aims to provide a technological contribution that supports aviation safety and benefits society.

7. BUDGET AND JUSTIFICATION

- **Hardware Cost**

  Basic hardware resources such as a personal computer or laptop with sufficient processing power and memory are required for data preprocessing, machine learning model development, and system implementation. The project will mainly utilize existing computing devices; however, minor hardware upgrades such as additional storage (SSD or external drive), RAM expansion, or other small accessories may be required to improve processing performance during model training and data analysis.

  Estimated Cost: Rs. 15,000

- **Software and Cloud Services**

  The project will primarily use open-source software tools such as Python, TensorFlow, Scikit-learn, and data visualization libraries for machine learning model development and weather data analysis. In addition, cloud services such as Amazon Web Services (AWS) may be used to store meteorological datasets and perform

computational experiments for model training and testing. Since most software tools are open-source, the cost mainly covers cloud service usage and minor development tools.

Estimated Cost: RS. 20000

- **Processing Resources**

  Additional computational resources may be required to efficiently process large meteorological datasets and run machine learning algorithms. These resources will support data preprocessing, model training, and evaluation tasks required for the aviation weather forecasting system.

  Estimated Cost: Rs. 10,000

Budget Justification

This study is conducted as an academic research project with limited financial requirements. Most of the required resources, including computing devices and development tools, are already available and open-source software will be used for machine learning model development and data visualization. Cloud services such as AWS will only be used on a limited scale for storing meteorological datasets and running computational experiments. Since this research is conducted for academic and experimental purposes and is not intended as a commercial system to be sold to government authorities, the budget allocation remains minimal. The estimated costs mainly support basic computational resources required to conduct and evaluate the proposed aviation weather forecasting system.

8. WORK BREAKDOWN STRUCTURE (WBS)

Task ID	Task Description	Timeline	Deliverable
1	Literature Review & Problem Identification (aviation weather + ML models)	March	Identified research gap + background study + Approved project proposal
2	Proposal Preparation & Submission	April	Cleaned & structured dataset
3	Data Collection (Airport Meteorological Data – CMB)	May – June	Raw historical weather dataset
4	Data Preprocessing (cleaning, normalization, time-series formatting)	July	Final feature set for model
5	Feature Engineering (Temp, Humidity, Pressure interactions, lag features)	July	Ready to Research 50%
6	Development of Coupled Prediction Model (LSTM + Multuple Regression Hybrid)	August – September	Trained hybrid prediction model
7	Model Evaluation & Forecast Verification (using METAR & TAF)	October	Accuracy results + verification report
8	Weather Visualization Development (heatmaps, wind maps, UI outputs)	June – September	Visualization module/dashboard
9	System Testing & Result Validation	October	Tested and validated system outputs
10	Final Report & Website Development	November	Completed system + documentation
11	Final Presentation & Viva	December	Project presentation
12	Research Paper Submission	December	Submitted research paper
13	Research Paper Publication Evidence	End of December	Publication proof

Table 2- Work breakdown table

9. GANTT CHART

Task	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Literature Review & Problem identification										
Proposal Preparation & Submission										
Data Collection (Airport Meteorological Data)										
Data Preprocessing & Feature Engineering										
Development of Coupled Prediction Model										
Model Evaluation & Forecast verification										
Weather Visualization Development										
System Testing & Result Validation										
Final Report & Website Development										
Final Presentation & Viva										
Research Paper Submission										
Research Paper Publication Evidence										

Table 3-project Implementation Timeline (Gantt Chart)

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11.APPENDICES

11.1 Appendix A: Survey Instruments

This research does not involve surveys or questionnaires because the study focuses on meteorological data analysis and machine learning–based weather prediction. Therefore, no survey instruments are required for this project.

11.2 Appendix B: Preliminary UI Mockup

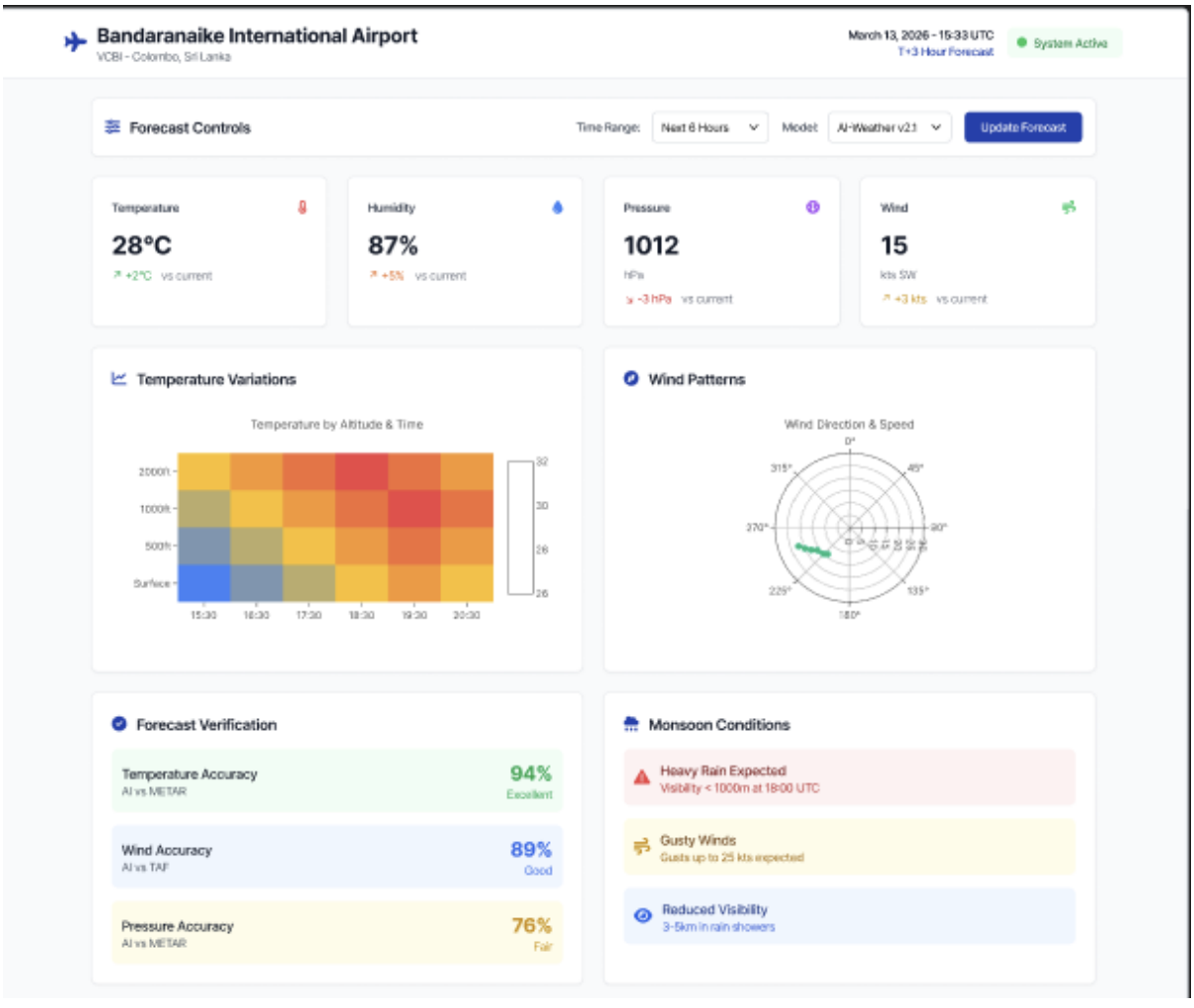


Figure 4 - main Ui with Temperature Variants

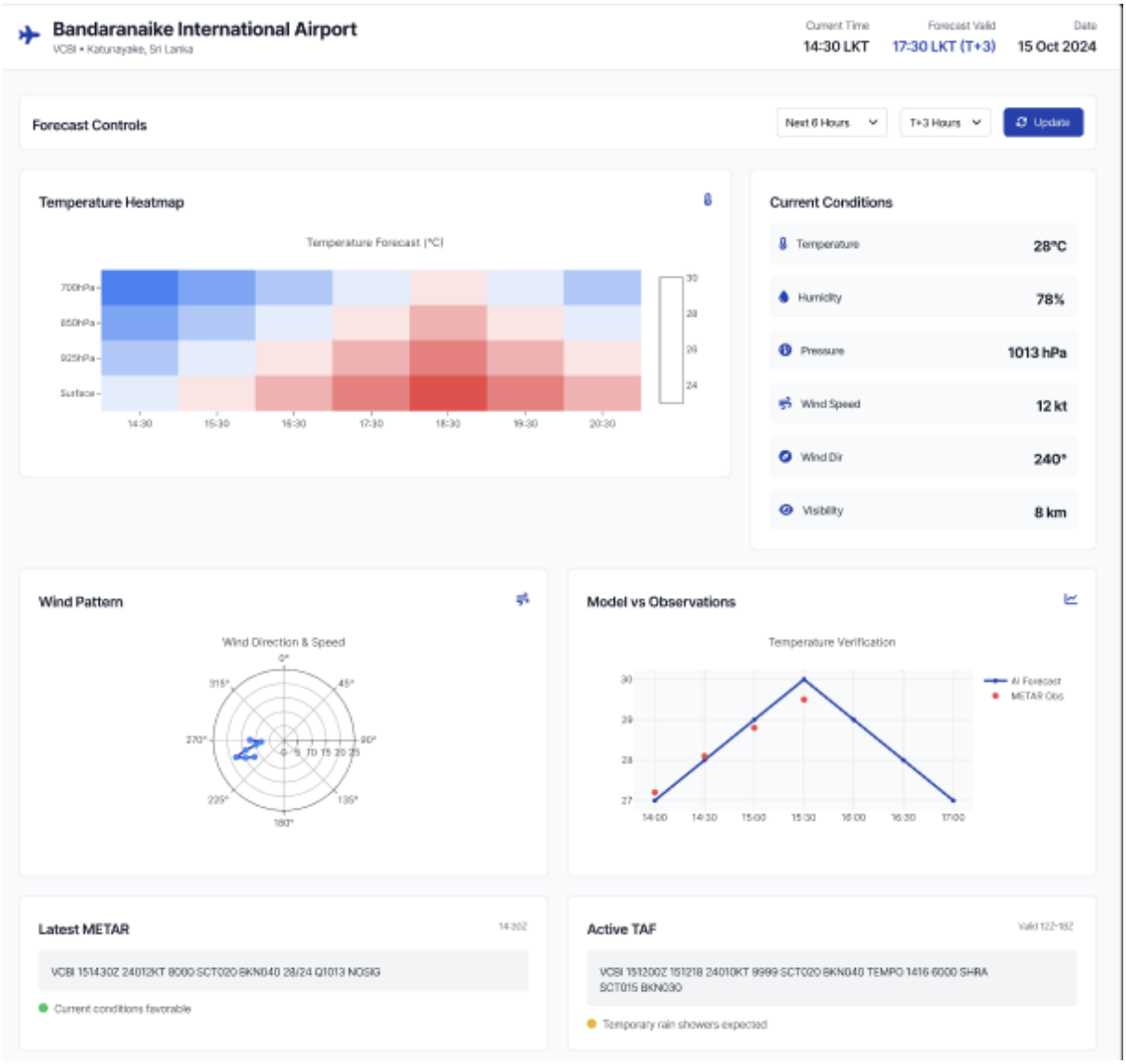


Figure 5 - Heatmap and wind pattern UI

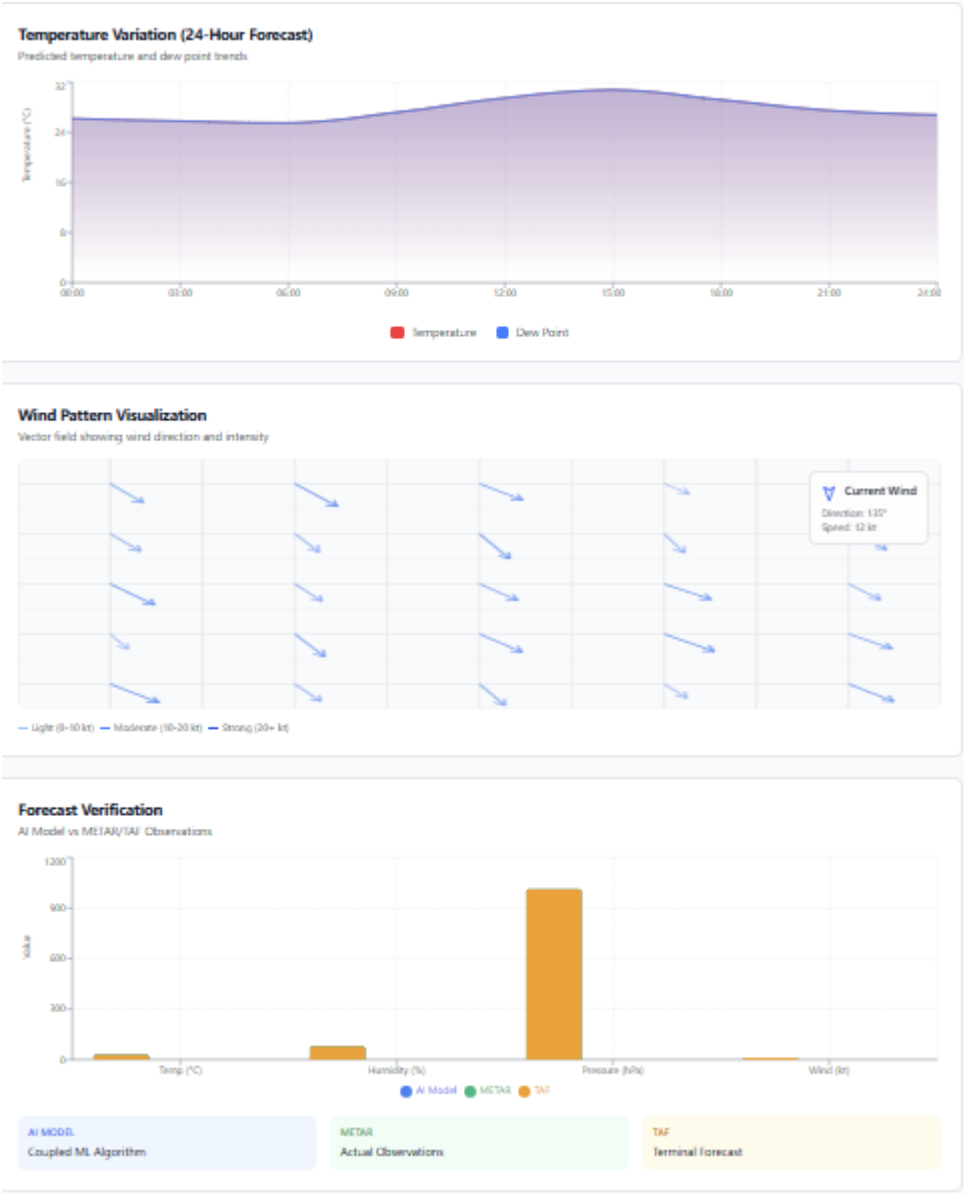


Figure 6 - Temperature and Forecast Verification

11.3 Appendix C: Risk Analysis

Risk	Description	Mitigation Strategy
Limited data availability	Historical weather data may be incomplete or missing values	Use data preprocessing and cleaning techniques to handle missing values
Model prediction errors	Machine learning model may produce inaccurate predictions	Evaluate model using validation metrics and compare with METAR / TAF data
Computational limitations	Training machine learning models may require high computing power	Use optimized algorithms and cloud computing resources if needed
Data inconsistency	Weather datasets may have different formats or measurement intervals	Standardize and normalize data during preprocessing
Overfitting of model	The model may perform well on training data but poorly on new data	Use cross-validation and testing datasets to evaluate model performance
Data quality issues	Sensor errors or inaccurate measurements may affect prediction quality	Apply data filtering and anomaly detection techniques
Model generalization limitations	The model may only work well for one location or dataset	Train and test the model using multiple time periods of data
System performance issues	Large datasets may slow down system processing	Use efficient data processing methods and optimized ML libraries

Table 4- Risk Analysis table

11.4 Appendix A: Survey Instruments

This research does not involve human subjects or personal data. The system uses meteorological data obtained through discussions and collaboration with the Meteorology Department and is supplemented with publicly available aviation weather observations.

Ethical considerations include:

Proper acknowledgment and citation of all meteorological datasets provided by the Meteorology Department and public sources. Ensuring data integrity, accuracy, and responsible handling of all datasets. Using the system strictly for decision-support and research purposes, not for direct operational control. Avoiding misuse of prediction results in safety-critical aviation environments. Maintaining transparency and respecting the intellectual contribution of the Meteorology Department in providing the data.